LSSO: Learnable Low-Rank Sylvester Solves for Efficient Bidirectional Encoders

Anonymous Authors

Abstract

We introduce LSSO, a learnable low-rank global-solve token mixer for bidirectional encoders. Instead of explicit pairwise value routing, LSSO constructs an input-dependent positive-shifted low-rank relation field and solves

$$(\mu I + \gamma U(X)U(X)^\top)Y = C(X).$$

The current operator is a Sylvester-style solve, corresponding to the left-sided special case of the more general equation $AY + YB = C$. We show that the operator is well posed under positive parameterization, admits an efficient Woodbury implementation, has a bounded fixed-U spectral response, and is equivalent to minimizing a strongly convex low-rank correction energy. Experiments on retrieval, MS MARCO to BEIR transfer, vision encoders, ablations, rank pruning, and sequence scaling show that LSSO can provide competitive bidirectional encoder performance with substantially lower mixer-only MACs. Boundary studies further clarify that LSSO is not an attention approximation and is not yet a universal replacement for attention in all generative settings.

1 Introduction

Self-attention [21] has become the default token mixer for bidirectional encoders. Given query, key, and value projections, it produces

$$Y_{\text{attn}} = \operatorname{softmax}(QK^\top/\sqrt{d})V,$$

which can be interpreted as input-dependent pairwise routing. This routing view is expressive, but the score and value-routing terms scale quadratically with sequence length. Many efficient attention methods reduce this cost by approximating or sparsifying the attention matrix. LSSO takes a different route: it replaces pairwise value routing with an input-dependent global linear solve.

For an input sequence $X \in \mathbb{R}^{B\times N\times D}$, LSSO learns a low-rank relation feature $U(X)$ and a content state $C(X)$. For each head it solves

$$(\mu I_N + \gamma UU^\top)Y = C, \qquad \mu > 0, \quad \gamma \geq 0,$$

and projects the result back to the model dimension. The matrix $UU^\top$ is positive semidefinite, while individual similarities $u_i^\top u_j$ may be positive or negative. Thus the system defines a stable global relation field without requiring an explicit dense routing matrix.

The guiding interpretation is:

$$\begin{aligned} \text{LSSO} = {} & \text{isotropic content pass-through} \\ & - \text{low-rank global correction.} \end{aligned}$$

This differs from attention: attention learns a row-stochastic routing matrix, whereas LSSO learns a positive-definite inverse filter.

Our contributions are:

1. We propose LSSO, a learnable Sylvester-style low-rank global solve operator for bidirectional encoder token mixing.
2. We derive a Woodbury implementation that reduces the solve to an $r \times r$ system, where r is the rank.
3. We prove fixed-U well-posedness, spectral boundedness, conditioning control, and an energy minimization equivalence.
4. We validate LSSO on retrieval, transfer, vision encoders, ablations, rank pruning, and long-sequence scaling.
5. We state the limitations explicitly: LSSO is a bidirectional encoder mixer, not an attention approximation, a causal decoder drop-in, or a universal diffusion backbone.

2 Related Work

Efficient attention. Linformer [23], Performer [4], Nyströmformer [25], Longformer [2], and BigBird [27] reduce the cost of self-attention through low-rank projections, kernel approximations, landmark approximations, or sparse attention patterns. These methods are best understood as efficient attention or attention-approximation mechanisms. LSSO does not attempt to approximate the softmax attention matrix; it defines a different input-dependent inverse-filter token mixer.

Token mixers beyond attention. MLP-Mixer [19], gMLP [13], FNet [12], Hyena [17], and state-space models such as Mamba [10] show that token mixing need not be implemented as dense attention. LSSO adds another point in this design space: an implicit low-rank global solve specialized for non-causal encoders.

Implicit layers and learned solves. Deep equilibrium models [1], implicit layers, graph smoothing, and regularized least-squares methods use linear or nonlinear solves as model components. LSSO is much simpler than a general equilibrium layer: it performs one closed-form, positive-shifted low-rank correction whose matrix is generated from the input.

3 Method

3.1 Encoder Block

We use LSSO in the token-mixer position of a pre-normalized encoder block:

$$X' = X + \mathrm{LSSO}(\mathrm{LN}(X)), \qquad X_{\text{out}} = X' + \mathrm{MLP}(\mathrm{LN}(X')).$$

All other components are kept unchanged in controlled comparisons: layer normalization, residual paths, feed-forward layers, positional embeddings, and task heads.

3.2 LSSO Definition

For each head, let

$$U = XW_u \in \mathbb{R}^{B\times N\times r}, \qquad C = XW_c \in \mathbb{R}^{B\times N\times d_h}.$$

We optionally RMS-normalize U along its rank dimension and solve

$$(\mu I_N + \gamma UU^\top)Y = C, \tag{1}$$

where

$$\mu = \mathrm{softplus}(\theta_\mu) + \epsilon, \qquad \gamma = \gamma_{\max}\sigma(\theta_\gamma).$$

The default experiments use $\gamma_{\max} = 0.3$ and $\theta_\gamma = -4$, which gives an initial γ/μ of roughly 7.8×10^{-3} when $\theta_\mu = 0$. Smaller settings such as $\gamma_{\max} = 0.1, \theta_\gamma = -6$ make the global correction nearly vanish at initialization and were empirically too conservative for the main runs.

The head outputs are concatenated and projected:

$$\mathrm{LSSO}(X) = YW_o.$$

3.3 Woodbury Implementation

Directly inverting the $N\times N$ matrix in Eq. 1 is unnecessary. By the Woodbury identity [24],

$$Y = \mu^{-1}C - \mu^{-1}U\left(I_r + \gamma\mu^{-1}U^\top U\right)^{-1}\left(\gamma\mu^{-1}U^\top C\right). \tag{2}$$

This form is continuous at $\gamma = 0$ and only solves an $r\times r$ system. The core solve terms cost

$$O(Nr^2 + Nrd_h + r^3)$$

per head. Including projections, the mixer-only MAC formula used in our tables is

$$ND(Hr + D) + ND^2 + HNr^2 + 2NrD + Hr^3.$$

For MHA we count QKV projections, attention scores, attention-value product, and output projection:

$$4ND^2 + 2N^2D.$$

4 Theoretical Properties

The following results analyze the operator for fixed U. They do not claim global convergence or stability of an entire trained deep network; those are empirical questions beyond the scope of this operator analysis.

Proposition 1 (Well-posedness)**.** *Let $A = \mu I_N + \gamma UU^\top$ with $\mu > 0$ and $\gamma \geq 0$. Then A is symmetric positive definite. Therefore, for any C, the system $AY = C$ has a unique solution.*

Proof. For any nonzero z,

$$z^\top Az = \mu\|z\|_2^2 + \gamma\|U^\top z\|_2^2 \geq \mu\|z\|_2^2 > 0.$$

□

Proposition 2 (Spectral shrinkage)**.** *Let $UU^\top = Q\Lambda Q^\top$ be an eigendecomposition over the nonzero eigen-directions of $UU^\top$, with $\Lambda = \mathrm{diag}(\lambda_1,\ldots,\lambda_k)$ and $\lambda_i > 0$. Let $P_\perp = I - QQ^\top$. Then*

$$A^{-1} = Q\,\mathrm{diag}\left(\frac{1}{\mu + \gamma\lambda_i}\right)Q^\top + \mu^{-1}P_\perp.$$

Thus LSSO applies a learnable inverse spectral filter along the span of U and isotropic scaling on the orthogonal complement. In this sense it is a data-dependent inverse filter rather than a row-stochastic routing matrix.

Proposition 3 (Bounded fixed-U operator norm)**.** *For fixed U,*

$$\|\,\mathrm{LSSO}(C,U)\|_F \leq \mu^{-1}\|C\|_F.$$

With an output projection,

$$\|W_o\,\mathrm{LSSO}(C,U)\|_F \leq \|W_o\|_2\mu^{-1}\|C\|_F.$$

Proof. Since $A \succeq \mu I$, $\lambda_{\min}(A) \geq \mu$ and $\|A^{-1}\|_2 \leq \mu^{-1}$. The Frobenius bound follows from $Y = A^{-1}C$. □

Proposition 4 (Conditioning control)**.** *For $A = \mu I + \gamma UU^\top$,*

$$\kappa_2(A) \leq 1 + \frac{\gamma}{\mu}\|U\|_2^2.$$

This explains the diagnostics used in our experiments. The ratio γ/μ directly scales the possible condition-number increase; effective rank measures how much of the rank subspace is used; and correction ratio measures how strongly the low-rank correction participates in the output. RMS normalization of U helps control the scale of U, although it is not by itself a dimension-free spectral-norm guarantee.

Theorem 1 (Energy minimization equivalence)**.** *For $\mu > 0$ and $\gamma \geq 0$, the LSSO output is the unique minimizer of*

$$\mathcal{E}(Y) = \frac{\mu}{2}\|Y\|_F^2 + \frac{\gamma}{2}\|U^\top Y\|_F^2 - \langle C, Y\rangle.$$

Proof. The Hessian of $\mathcal{E}$ is $\mu I + \gamma UU^\top$, which is positive definite by the well-posedness proposition. Setting the gradient to zero gives $(\mu I + \gamma UU^\top)Y = C$. □

Equivalently, up to an additive constant,

$$\mathcal{E}(Y) = \frac{\mu}{2} \left\| Y - \mu^{-1} C \right\|_F^2 + \frac{\gamma}{2} \| U^\top Y \|_F^2.$$

The content projection proposes the local state $\mu^{-1}C$; the learned low-rank term applies a global correction.

Proposition 5 (No-global degeneration)**.** *If $\gamma = 0$, then*

$$\mathrm{LSSO}(C, U) = \mu^{-1} C.$$

No token mixing occurs except for pointwise projections.

5 Experiments

5.1 Setup

All main retrieval and vision comparisons use randomly initialized encoders. The retrieval backbone follows a BERT-style bidirectional encoder layout [8]; tokenizers may be reused, but encoder weights are trained from scratch. Models are optimized with AdamW [14]. LSSO uses $\gamma_{\max} = 0.3$ and $\theta_\gamma = -4$ unless stated otherwise. Tables report mixer-only MACs to isolate the cost of replacing the token mixer; FFN cost is intentionally not folded into the main MAC column.

5.2 Retrieval Main Results

Table 1 reports 3-seed retrieval results on FIQA [15], NFCorpus [3], and SciFact [22], using their BEIR retrieval formulations [18]. The baseline is a BERT-style encoder with MHA. Nyströmformer uses the official attention module. LSSO-r16 and LSSO-r32 keep the same model dimension, depth, heads, optimizer, and training schedule.

Table 1: Retrieval main results. Models are randomly initialized BERT-style encoders with $D = 256$, depth 8, heads 8, and document length 512. MACs are mixer-only document-side MACs.

Dataset	Model	Params (M)	Mixer MACs (G)	R@10	MRR@10
FIQA	MHA	14.33	2.147	0.2145 ± 0.0135	0.0987 ± 0.0032
FIQA	Nyströmformer	14.33	1.301	0.2428 ± 0.0039	0.1190 ± 0.0071
FIQA	LSSO-r16	13.54	0.713	0.2258 ± 0.0125	0.1126 ± 0.0061
FIQA	LSSO-r32	13.80	0.908	0.2387 ± 0.0183	0.1150 ± 0.0035
NFCorpus	MHA	14.33	2.147	0.5439 ± 0.0187	0.3809 ± 0.0145
NFCorpus	Nyströmformer	14.33	1.301	0.5841 ± 0.0125	0.4323 ± 0.0132
NFCorpus	LSSO-r16	13.54	0.713	0.5686 ± 0.0125	0.4121 ± 0.0036
NFCorpus	LSSO-r32	13.80	0.908	0.5841 ± 0.0036	0.4333 ± 0.0220
SciFact	MHA	14.33	2.147	0.6789 ± 0.0190	0.5994 ± 0.0138
SciFact	Nyströmformer	14.33	1.301	0.7267 ± 0.0133	0.6313 ± 0.0033
SciFact	LSSO-r16	13.54	0.713	0.7022 ± 0.0107	0.6214 ± 0.0119
SciFact	LSSO-r32	13.80	0.908	0.7089 ± 0.0117	0.6247 ± 0.0080

LSSO-r16 reduces mixer MACs by 66.8% relative to MHA, while LSSO-r32 reduces them by 57.7%. Both retain or improve MHA-level retrieval quality across the three datasets. Nyströmformer is a strong retrieval baseline in this setup, but its observed throughput is lower than both MHA and LSSO in our runs.

5.3 MS MARCO to BEIR Transfer

To test whether the result is limited to small supervised retrieval datasets, we pretrain on MS MARCO [16] and evaluate zero-shot on BEIR-style tasks [18]. This setting is stricter than the per-dataset retrieval runs above: the model is trained once on a large web retrieval source and is then evaluated without task-specific finetuning on finance, biomedical, scientific fact-checking, argument retrieval, and COVID retrieval datasets. It therefore probes whether the token mixer learns a general bidirectional retrieval encoder rather than a dataset-specific scoring shortcut.

All four models use the same randomly initialized BERT-style architecture, MS MARCO training data, tokenizer, pooling, optimizer, three seeds, and document length 512. Only the token mixer changes. Macro averages over FIQA, NFCorpus, SciFact, ArguAna, and TREC-COVID are shown in Table 2; per-dataset nDCG@10 is shown in Table 3.

Table 2: MS MARCO to BEIR transfer macro average over five datasets.

Model	MACs (G)	nDCG@10	Recall@100
MHA	2.147	0.22945	0.36061
Nyströmformer	1.270	0.20523	0.33456
LSSO-r16	0.713	0.22973	0.35889
LSSO-r32	0.908	0.22940	0.36367

The LSSO models preserve MHA-level transfer under substantially lower mixer cost. LSSO-r16 nearly matches the MHA macro nDCG@10 (0.22973 vs. 0.22945) while using 66.8% fewer mixer MACs. LSSO-r32 has the strongest macro Recall@100 among the four models (0.36367) while still using 57.7% fewer mixer MACs. In this setting the official Nyströmformer baseline is cheaper than MHA but trails both MHA and LSSO on the macro averages, suggesting that the low-rank solve is not merely a cheaper attention approximation but a competitive transfer-oriented encoder bias.

Table 3: Zero-shot BEIR transfer after MS MARCO pretraining: per-dataset nDCG@10 over three seeds.

Dataset	MHA	Nyström	LSSO-r16	LSSO-r32
ArguAna	0.20259	0.18247	0.19896	0.19783
FIQA	0.07372	0.07253	0.07322	0.07686
NFCorpus	0.18565	0.16607	0.18015	0.18704
SciFact	0.29838	0.25846	0.30881	0.33148
TREC-COVID	0.38694	0.34661	0.38749	0.35382
Macro avg.	0.22945	0.20523	0.22973	0.22940

The per-dataset pattern is mixed, which is expected for zero-shot BEIR: MHA is best on ArguAna and TREC-COVID, LSSO-r32 is best on FIQA, NFCorpus, and SciFact, and LSSO-r16 is essentially tied with MHA on the macro average. The important result is not that one LSSO rank dominates every target dataset, but that both LSSO ranks remain at MHA-level transfer quality while operating at a much lower mixer cost.

5.4 Ablations and Diagnostics

Table 4 aligns the empirical ablations with the theory. The no-global variant sets $\gamma = 0$, reducing the operator to $\mu^{-1}C$ and removing token mixing. Removing RMS normalization weakens the global correction, especially on SciFact. Smaller ranks expose the capacity of the learned relation subspace.

Table 4: Retrieval ablations. Values are mean $\pm$ standard deviation over three seeds.

Dataset	Variant	R@10	MRR@10	γ/μ	Correction ratio
FIQA	no-global r32	0.2052 ± 0.0137	0.0911 ± 0.0019	0.0000	0.0000
FIQA	fixed μ/γ r32	0.2428 ± 0.0126	0.1136 ± 0.0028	0.0078	0.2130
FIQA	no U RMS norm r32	0.2160 ± 0.0101	0.1007 ± 0.0008	0.0085	0.2050
FIQA	full r8	0.2088 ± 0.0208	0.0923 ± 0.0139	0.0100	0.1059
FIQA	full r4	0.2042 ± 0.0045	0.0979 ± 0.0036	0.0109	0.0742
SciFact	no-global r32	0.6767 ± 0.0120	0.5811 ± 0.0058	0.0000	0.0000
SciFact	fixed μ/γ r32	0.7089 ± 0.0102	0.6088 ± 0.0066	0.0078	0.2631
SciFact	no U RMS norm r32	0.6933 ± 0.0133	0.5972 ± 0.0150	0.0078	0.0471
SciFact	full r8	0.7067 ± 0.0173	0.6118 ± 0.0064	0.0078	0.1149
SciFact	full r4	0.7022 ± 0.0069	0.6146 ± 0.0107	0.0078	0.0766

Figure 1 plots trained layer-wise diagnostics. All tasks show nonzero correction ratios, confirming that the trained operator does not collapse to the local projection. Effective rank is task dependent: SciFact uses much more of the available rank-32 subspace than FIQA, NFCorpus, or CIFAR-100.

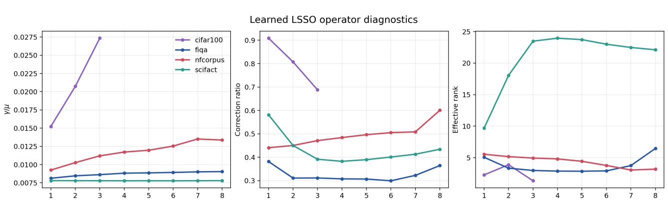

Figure 1: Layer-wise LSSO diagnostics from trained LSSO-r32 checkpoints.

5.5 Sequence Scaling

We benchmark single-layer token mixers at sequence lengths from 128 to 2048 with batch size 8, $D = 256$, 8 heads, bf16 autocast, and no optimizer update. Figure 2 shows forward latency, forward+backward latency, incremental peak memory, and theoretical mixer MACs.

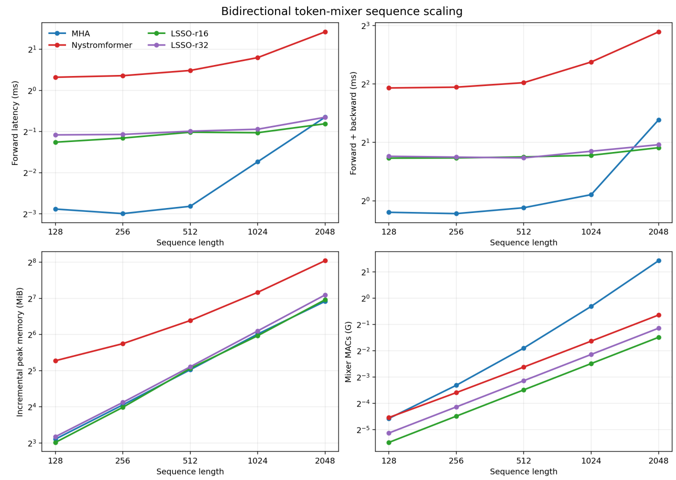

Figure 2: Sequence scaling benchmark on an NVIDIA RTX 5070 Ti. MACs are theoretical per-sample single-layer mixer MACs; latency and memory use batch 8.

At $N = 2048$, LSSO-r16 uses 0.357G mixer MACs compared with 2.684G for MHA, an 86.7% reduction, and reduces forward+backward latency from 2.61 ms to 1.88 ms. LSSO-r32 uses 0.453G MACs and takes 1.95 ms. Peak memory is closer to optimized MHA than the dense $O(N^2)$ formula would suggest because current PyTorch MHA uses memory-efficient Flash/SDPA-style kernels [6]. Thus this benchmark supports the arithmetic scaling and long-sequence latency claim, not a universal memory-savings claim.

5.6 Vision and Diffusion Boundary Results

Table 5 reports ViT-style [9] vision encoder results on CIFAR-100 [11] and an ImageNet subset [7]. The controlled vision runs use RandAugment [5], Mixup [28], and CutMix [26], following data-efficient vision-transformer training practice [20]. On CIFAR-100, LSSO gives a cost-accuracy trade-off but does not dominate the official Nyströmformer baseline. On ImageNet-100, MHA remains stronger at one seed, while LSSO reduces mixer MACs substantially. These results support LSSO as a general encoder mixer candidate rather than a universal replacement.

Table 5: Vision encoder results. CIFAR-100 is 3 seeds; ImageNet-100 is one seed.

Dataset	Model	MACs (G)	Top-1	Top-5
CIFAR-100	MHA	0.0665	55.40 ± 0.27	–
CIFAR-100	Nyströmformer	0.0389	59.98 ± 0.58	–
CIFAR-100	LSSO-r16	0.0249	54.79 ± 0.14	–
CIFAR-100	LSSO-r32	0.0385	55.38 ± 0.18	–
ImageNet-100	MHA	0.520	82.26	95.54
ImageNet-100	LSSO-r16	0.137	80.54	94.88
ImageNet-100	LSSO-r32	0.174	80.58	94.76

We also ran a latent diffusion boundary experiment on ImageNet-100 latent tokens. It reports noise-prediction validation MSE only, without FID/KID or sample-quality claims. Pure LSSO retains much of the MHA optimization quality with lower theoretical mixer MACs, but it is not yet a direct replacement for attention in diffusion generation.

6 Discussion

What LSSO is. LSSO is a low-rank global solve, a stable implicit correction operator for fixed U, and a bidirectional encoder token mixer.

What LSSO is not. LSSO is not an attention approximation. It is not currently a causal decoder-only replacement, because causal masking destroys the simple low-rank Woodbury structure. It is also not yet a universal diffusion backbone.

Why retrieval is a natural fit. Retrieval encoders require global bidirectional semantic compression rather than precise causal next-token routing. A low-rank global correction can capture document-level structure while avoiding explicit all-pairs value routing.

7 Limitations

First, the theory analyzes fixed-U operator properties. It does not prove stability of full end-to-end training dynamics. Second, LSSO is designed for bidirectional encoders; causal and autoregressive extensions require new prefix-restricted solves or cached global states. Third, rank is a real capacity knob: some tasks use only a compact subspace, while others, such as SciFact in our diagnostics, use much more of rank 32. Fourth, vision and diffusion experiments show useful cost-accuracy trade-offs but not universal dominance over attention or strong attention alternatives.

8 Conclusion

We introduced LSSO, a learnable low-rank Sylvester-style solve operator for efficient bidirectional token mixing. The operator is well posed, efficiently computable through Woodbury, spectrally bounded for fixed relation features, conditioned by γ/μ and the scale of U, and equivalent to solving a strongly convex global correction energy. Experiments show strong retrieval and transfer performance under lower mixer-only cost, while boundary studies clarify where LSSO is and is not currently effective.

References

[1] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models. *Advances in Neural Information Processing Systems*, 2019.

[2] Iz Beltagy, Matthew E. Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.

[3] Vera Boteva, Demian Gholipour Ghalandari, Artem Sokolov, and Stefan Riezler. A full-text learning to rank dataset for medical information retrieval. In *Advances in Information Retrieval*, pages 716–722, 2016. doi: 10.1007/978-3-319-30671-1_58.

[4] Krzysztof Choromanski, Valerii Likhosherstov, David Dohan, Xingyou Song, Andreea Gane, Tamas Sarlos, Peter Hawkins, Jared Davis, Afroz Mohiuddin, Lukasz Kaiser, David Belanger, Lucy Colwell, and Adrian Weller. Rethinking attention with performers. *International Conference on Learning Representations*, 2021.

[5] Ekin D. Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V. Le. Randaugment: Practical automated data augmentation with a reduced search space. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 702–703, 2020. doi: 10.1109/CVPRW50498.2020.00359.

[6] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with io-awareness. *Advances in Neural Information Processing Systems*, 35, 2022.

[7] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009. doi: 10.1109/CVPR.2009.5206848.

[8] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019. doi: 10.18653/v1/N19-1423.

[9] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations*, 2021.

[10] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.

[11] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

[12] James Lee-Thorp, Joshua Ainslie, Ilya Eckstein, and Santiago Ontiveros. Fnet: Mixing tokens with fourier transforms. *arXiv preprint arXiv:2105.03824*, 2021.

[13] Hanxiao Liu, Zihang Dai, David R. So, and Quoc V. Le. Pay attention to mlps. *Advances in Neural Information Processing Systems*, 2021.

[14] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

[15] Macedo Maia, Siegfried Handschuh, André Freitas, Brian Davis, Ross McDermott, Manel Zarrouk, and Alexandra Balahur. WWW'18 open challenge: Financial opinion mining and question answering. In *Companion Proceedings of The Web Conference 2018*, pages 1941–1942, 2018.

[16] Tri Nguyen, Mir Rosenberg, Xia Song, Jianfeng Gao, Saurabh Tiwary, Rangan Majumder, and Li Deng. MS MARCO: A human generated machine reading comprehension dataset. *arXiv preprint arXiv:1611.09268*, 2016.

[17] Michael Poli, Stefano Massaroli, Eric Nguyen, Daniel Y. Fu, Tri Dao, Stephen Baccus, Yoshua Bengio, Stefano Ermon, and Christopher Ré. Hyena hierarchy: Towards larger convolutional language models. *International Conference on Machine Learning*, 2023.

[18] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. Beir: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. *Advances in Neural Information Processing Systems*, 2021.

[19] Ilya O. Tolstikhin, Neil Houlsby, Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Thomas Unterthiner, Jessica Yung, Andreas Steiner, Daniel Keysers, Jakob Uszkoreit, Mario Lucic, and Alexey Dosovitskiy. Mlp-mixer: An all-mlp architecture for vision. *Advances in Neural Information Processing Systems*, 2021.

[20] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers and distillation through attention. In *International Conference on Machine Learning*, pages 10347–10357, 2021.

[21] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.

[22] David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. Fact or fiction: Verifying scientific claims. In *Proceedings of EMNLP*, pages 7534–7550, 2020. doi: 10.18653/v1/2020.emnlp-main.609.

[23] Sinong Wang, Belinda Z. Li, Madian Khabsa, Han Fang, and Hao Ma. Linformer: Self-attention with linear complexity. *arXiv preprint arXiv:2006.04768*, 2020.

[24] Max A. Woodbury. Inverting modified matrices. Technical Report Memorandum Report 42, Statistical Research Group, Princeton University, 1950.

[25] Yunyang Xiong, Zhanpeng Zeng, Rudrasis Chakraborty, Mingxing Tan, Glenn Fung, Yin Li, and Vikas Singh. Nyströmformer: A nyström-based algorithm for approximating self-attention. *AAAI Conference on Artificial Intelligence*, 2021.

[26] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. Cutmix: Regularization strategy to train strong classifiers with localizable features. In *IEEE/CVF International Conference on Computer Vision*, pages 6023–6032, 2019. doi: 10.1109/ICCV.2019.00612.

[27] Manzil Zaheer, Guru Guruganesh, Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontiveros, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, and Amr Ahmed. Big bird: Transformers for longer sequences. *Advances in Neural Information Processing Systems*, 2020.

[28] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. In *International Conference on Learning Representations*, 2018.

A Full Woodbury Derivation

The Woodbury identity gives

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}.$$

Set $A = \mu I_N$, $C = \gamma I_r$, and $V = U^\top$:

$$(\mu I + \gamma UU^\top)^{-1} = \mu^{-1}I - \mu^{-1}U(\gamma^{-1}I + \mu^{-1}U^\top U)^{-1}U^\top \mu^{-1}.$$

For implementation it is preferable to avoid γ^{-1}, so we rewrite the solution as

$$Y = \mu^{-1}C - \mu^{-1}U(I + \gamma\mu^{-1}U^\top U)^{-1}(\gamma\mu^{-1}U^\top C).$$

This expression remains valid at $\gamma = 0$.

B Implementation Notes

The PyTorch implementation computes $U^\top U$ and $U^\top C$ in batched form, solves the small $r \times r$ system in float32 for numerical stability under fp16/bf16 autocast, and casts the result back to the activation dtype. Diagnostics record γ/μ, effective rank of $U^\top U$, and the ratio between the Woodbury correction and the local $\mu^{-1}C$ term.